

Executive Summary

This report proposes an intensive **3-day study plan** (4–5 hours/day) designed to cover core STEM/mathematics topics. Because the subject was unspecified, we assume a **math-focused curriculum** (e.g. Precalculus-level topics) but the schedule is easily adapted to other quantitative fields. Each day consists of **time-blocked study sessions** (~30–60 min each) with short breaks, plus review periods. Research shows that structuring study into focused blocks with planned breaks improves focus and learning: for example, using the Pomodoro technique (25–60 min work, 5–10 min break) helps maintain concentration [\[17†L156-L163\]](#) , and taking **purposeful breaks** (even 5–20 min) refreshes the brain and boosts productivity [\[40†L203-L208\]](#) [\[21†L202-L210\]](#) . We also double our task estimates to avoid under-scheduling [\[17†L159-L163\]](#) . Active learning (practice problems, self-testing) and spaced review are emphasized: retrieval practice (recalling or solving problems) strongly enhances retention [\[26†L61-L63\]](#) .

The plan includes: (1) a **detailed hourly schedule** for each day (with study blocks, breaks, and reviews); (2) a comprehensive **topic outline** covering functions, algebra, trigonometry, etc.; (3) **100+ practice problems** (with difficulty labels and time estimates), each with an answer and brief solution; (4) **print-friendly materials** (HTML/CSS snippets for printing schedules/problems); and (5) curated **references** supporting the strategy. All materials are structured with clear headings and tables for easy scanning.

3-Day Study Schedule

Below is the 3-day schedule. Each day is ~5 hours of study (plus breaks). Blocks range 45–60 min; breaks are 10–20 min. **Review/Q&A** blocks at the end of each day ensure active recall. The schedule is a template and can be adjusted to specific needs.

Time	Day 1 (Polynomials & Functions)	Day 2 (Exponents & Trigonometry)	Day 3 (Advanced Topics)
9:00–10:00	Quadratic & Polynomial Functions (<i>intro/examples</i>)	Exponential Functions (<i>growth, graphs</i>)	Complex Numbers & Polar Coordinates (<i>intro</i>)
10:00–10:15	Break	Break	Break
10:15–11:15	Rational Functions (<i>domain, asymptotes</i>)	Logarithmic Functions (<i>solve, properties</i>)	Parametric Equations (<i>e.g. projectile motion</i>)
11:15–11:30	Break	Break	Break
11:30–12:30	Function Composition / Inverses (<i>practice</i>)	Trig Basics: Unit Circle, Values	Vectors & Matrices (<i>operations</i>)
12:30–13:30	Lunch/Long Break (60 min)	Lunch/Long Break (60 min)	Lunch/Long Break (60 min)
13:30–14:30	Exponents & Logs (continuation)	Trig Graphs & Identities (<i>sum/diff</i>)	Conic Sections or Review of prior topics
14:30–14:45	Break	Break	Break
14:45–15:45	Review & Practice Problems	Review & Practice Problems	Review & Practice Problems

Each study block focuses on specific subtopics (see list below). Short breaks help maintain focus [\[17†L156-L163\]](#) [\[40†L203-L208\]](#) . For example, one can use the Pomodoro rhythm (25 min work + 5 min break) or longer blocks (50–60 min + 10–15 min break) based on attention span. A 20-minute break after ~90 minutes of work is shown to improve retention [\[21†L202-L210\]](#) . After each block, a brief review/Q&A consolidates learning (active recall) [\[26†L61-L63\]](#) .

timeline title 3-Day Study Plan (Daily Timeline) section Day 1 9:00–10:00 : Quadratics/Polynomials 10:15–11:15 : Rational Functions 11:30–12:30 : Function Composition/Inverses 13:30–14:30 : Exponential & Logarithmic Functions 14:45–15:45 : Review & Practice section Day 2 9:00–10:00 : Exponential Functions (cont.) 10:15–11:15 : Logarithmic Equations 11:30–12:30 : Trig Basics (Unit Circle, Values) 13:30–14:30 : Trig Graphs & Identities 14:45–15:45 : Review & Practice section Day 3 9:00–10:00 : Complex/Polar Coordinates 10:15–11:15 : Parametric Equations 11:30–12:30 : Vectors & Matrices 13:30–14:30 : Conics or Misc. Topics 14:45–15:45 : Review & Practice

Topics and Subtopics

Assumptions: We cover common STEM-quantitative topics (with a math emphasis). Key areas include:

- Functions and Graphs:** Definition of a function, domain/range, evaluating functions, graphing basic functions (lines, parabolas, etc.), transformations (shifts/reflections), piecewise functions.
- Polynomial Functions:** Quadratic functions (vertex, axis, roots), higher-degree polynomials, factoring, solving polynomial equations, graph features (intercepts, end-behavior).

- **Rational Functions:** Simplifying rational expressions, domain restrictions, asymptotes (vertical, horizontal), holes, graphing rational functions, solving rational equations/inequalities.
- **Exponential & Logarithmic:** Exponential growth/decay models, properties of exponents, defining logarithms, solving exponential and logarithmic equations, change-of-base, graph shapes.
- **Trigonometry:** Angle measure (degrees/radians), unit circle values, right-triangle definitions, graphs of sine/cosine/tangent (amplitude, period, phase shift), inverse trigonometric functions.
- **Trig Identities & Equations:** Fundamental identities (Pythagorean, sum/difference, double-angle), proving/simplifying expressions, solving trig equations (basic and using identities).
- **Composite & Inverse Functions:** Function composition and inversion, domain restrictions for inverses, graphical interpretations.
- **Complex Numbers:** Arithmetic with complex numbers, polar form, Euler's formula (optional), solving equations with complex roots.
- **Parametric & Polar Coordinates:** Converting between rectangular and polar forms, graphing polar equations, analyzing parametric motion (velocity, trajectory).
- **Vectors & Matrices:** Vector operations (addition, scalar multiplication, dot product, magnitude), basic 2×2 matrix multiplication and transformations (if needed).
- **Conic Sections & Analytic Geometry:** Equations/graphs of circles, ellipses, hyperbolas, parabolas; standard forms and key properties (foci, directrices).
- **Additional Topics:** Depending on need, topics like sequences/series, probability basics, or other domain-specific math.

These topics align with major STEM/math curricula. (For example, AP Precalculus covers Polynomials/Rationals, Exponentials/Logs, Trigonometry/Polar, etc.

【5†L159-L167】.)

100+ Practice Problems

Below are **100 practice problems** spanning the above topics. Each is tagged with Difficulty and estimated time. Answers and concise explanations follow each question. (All answers are given; try solving on paper first.)

Quadratics and Polynomial Functions

1. (**Easy, 2 min**) Find the roots of the equation $(x^2 - 5x + 6 = 0)$.

Answer: $(x=2)$ or $(x=3)$.

Explanation: Factor as $((x-2)(x-3)=0)$.

2. (**Easy, 2 min**) Expand and simplify: $((x+2)(x^2 - 2x + 3))$.

Answer: $(x^3 + 0x^2 - x + 6)$.

Explanation: Multiply: $(x^3 - 2x^2 + 3x + 2x^2 - 4x + 6 = x^3 - x + 6)$.

3. (**Medium, 4 min**) A quadratic function has vertex at $((-1, 2))$ and passes through $((0, -1))$. Write its formula.

Answer: $(f(x) = (x+1)^2 + 2 - 3(x+1))$ or $(f(x)=x^2+2x+1+2-3x-3)$ simplifies to $(x^2 - x + 0)$. Actually let's compute properly: Vertex form $(f(x)=a(x+1)^2+2)$. It passes $(0,-1)$: $(-1 = a(1)^2+2 \Rightarrow a=-3)$. So $(f(x)=-3(x+1)^2+2 = -3x^2-6x-1)$.

Answer: $(f(x) = -3(x+1)^2 + 2)$.

Explanation: Use vertex form $(a(x+1)^2+2)$. Plug $(0,-1)$ to find $(a = -3)$. Expand if needed.

4. (**Medium, 5 min**) A cubic polynomial $(p(x))$ has roots $(1, -2, 3)$ and leading coefficient 2. Find $(p(x))$.

Answer: $(p(x) = 2(x-1)(x+2)(x-3))$.

Explanation: Roots given, use factor form with leading coefficient 2.

5. (**Easy, 2 min**) For $(f(x)=4x^3-8x)$, find $(f(2))$.

Answer: $(f(2) = 4(8) - 8(2) = 32 - 16 = 16)$.

Explanation: Substitute $(x=2)$.

6. (**Hard, 8 min**) Sketch qualitatively the graph of $(y=(x-1)^2(x+2))$. Identify intercepts and end-behavior.

Answer: Zeros at $(x=1)$ (double root) and $(x=-2)$. As $(x \rightarrow +\infty)$, $(y \rightarrow +\infty)$; as $(x \rightarrow -\infty)$, $(y \rightarrow -\infty)$ (leading coefficient positive with odd degree). Intercepts: $((1,0), (-2,0))$. Graph touches at $(x=1)$ (flattened).

Explanation: A double root causes the parabola-like touch at $(x=1)$.

7. (**Medium, 4 min**) Solve $(x^4 - 5x^2 + 4 = 0)$.

Answer: Let $(u=x^2)$: $(u^2 - 5u + 4=0)$, so $(u=1)$ or $(u=4)$. Thus $(x^2=1)$ or (4) . Solutions: $(x = \pm 1, \pm 2)$.

Explanation: Use substitution for bi-quadratic.

8. **(Easy, 2 min)** Determine whether $f(x)=x^3-7x+6$ is even, odd, or neither.

Answer: Neither.

Explanation: Check $f(-x) = -x^3 + 7x + 6 \neq \pm f(x)$.

9. **(Medium, 5 min)** If $f(x) = ax^2 + bx + c$ has vertex $((h,k))$ and $f(0)=c$, express $f(x)$ in vertex form and find $f(0)$.

Answer: Vertex form: $f(x) = a(x-h)^2 + k$. Then $f(0)=ah^2 + k$.

Explanation: Standard vertex form substitution.

10. **(Medium, 5 min)** Given $(x^2 + y^2 - 4x + 6y - 12 = 0)$, identify its graph.

Answer: Circle. Completing square: $((x^2-4x+4)+(y^2+6y+9) = 12+4+9 = 25)$. So $((x-2)^2 + (y+3)^2 = 25)$. Center $(2,-3)$, radius 5.

Explanation: It's a circle centered at $(2,-3)$ of radius 5.

Rational Functions

11. **(Easy, 2 min)** Simplify $\frac{x^2-4}{x}$ for $x \neq 2$.

Answer: $(x+2)$.

Explanation: Factor numerator $((x-2)(x+2))$.

12. **(Easy, 2 min)** Find the vertical asymptote(s) of $f(x)=\frac{2x}{x^2-9}$.

Answer: $(x = 3)$ and $(x=-3)$.

Explanation: Denominator zero at (± 3) .

13. **(Medium, 4 min)** Determine the horizontal asymptote of $(y = \frac{3x^2 - x + 1}{2x^2 + 5x - 3})$.

Answer: $(y = \frac{3}{2})$.

Explanation: Leading coefficients ratio (same degree).

14. **(Medium, 4 min)** Solve $\frac{x+1}{x-2} = 3$.

Answer: $(x+1 = 3(x-2) \implies x+1 = 3x-6 \implies -2x=-7 \implies x=3.5)$.

Explanation: Cross-multiply (denominator nonzero).

15. **(Hard, 8 min)** For $f(x)=\frac{x^2-1}{x^2-4}$, find holes, vertical/horizontal asymptotes.

Answer: Factor $((x-1)(x+1)/(x-2)(x+2))$. No common factors, so no holes. Vertical asymptotes at $(x=2)$ and $(x=-2)$. Horizontal asymptote $(y=1)$ (degree equal, ratio 1/1).

Explanation: After factoring, check cancellations (none). Compare degrees.

16. **(Medium, 4 min)** Find all (x) so that $\frac{2x}{x^2-9}$ is undefined.

Answer: $(x=3)$ or $(x=-3)$.

Explanation: Denominator zero at (± 3) .

17. **(Medium, 5 min)** Simplify $\frac{1}{x+1} + \frac{2}{x-1}$.

Answer: $(\frac{(x-1) + 2(x+1)}{(x+1)(x-1)} = \frac{x-1+2x+2}{x^2-1} = \frac{3x+1}{x^2-1})$.

Explanation: Combine over common denominator.

18. **(Hard, 10 min)** Graph $(y = \frac{(x-2)(x+3)}{(x+1)^2})$. Identify intercepts, asymptotes, and any holes.

Answer: Numerator zero at $(x=2,-3)$, so x-intercepts there (if not canceled). Denominator zero at $(x=-1)$ (double). Vertical asymptote at $(x=-1)$. Horizontal asymptote: degree numerator 2, denom 2 $\Rightarrow$ ratio of leading coefficients $(1/1=1)$, so $(y=1)$. No hole (no cancellation). Y-intercept $(y = \frac{(-2)(3)}{(1)^2} = -6)$.

Explanation: Identify zeros and asymptotes by factoring.

19. **(Easy, 2 min)** Evaluate $(\lim_{x \rightarrow 2} \frac{x^2-4}{x-2})$.

Answer: 4.

Explanation: Simplify to $(\lim_{x \rightarrow 2} (x+2))$ as $(x \rightarrow 2)$, gives 4.

20. **(Medium, 5 min)** Solve the rational inequality $(\frac{x+2}{x-3} > 0)$.

Answer: Solution: $(x < -2)$ or $(x > 3)$. (Check sign table: numerator positive for $(x>3)$, denominator positive for $(x>-2)$, etc.)

Explanation: Determine sign by testing intervals around -2 and 3; exclude $(x=-2)$.

Exponential and Logarithmic Functions

21. **(Easy, 2 min)** Solve $(2^x = 16)$.

Answer: $(x=4)$.

Explanation: $(16=2^4)$.

22. (Medium, 5 min) Solve $(3e^{2x}) = 12$.

Answer: $(e^{2x} = 4)$, so $(2x = \ln 4)$, hence $(x = \frac{\ln 4}{2} = \ln 2)$.

Explanation: Divide by 3, take $\ln$.

23. (Medium, 4 min) Express $(\log_2(32))$ without a calculator.

Answer: (5).

Explanation: $(32 = 2^5)$.

24. (Medium, 4 min) Expand using log rules: $(\ln(\frac{e^{3x}\sqrt{z^2}}{y}))$.

Answer: $(3x + \frac{1}{2}\ln y - 2\ln z)$.

Explanation: $(\ln(A/B) = \ln A - \ln B)$; $(\sqrt{z^2} = z)$.

25. (Hard, 8 min) Solve for (x) : $(\log_3(x-1) + \log_3(x+2) = 2)$.

Answer: Combine logs: $(\log_3[(x-1)(x+2)] = 2)$. So $((x-1)(x+2) = 3^2 = 9)$. Solve $(x^2 + x - 2 = 9 \implies x^2 + x - 11 = 0)$. Solutions $(x = \frac{-1 \pm \sqrt{1+44}}{2}) = \frac{-1 \pm \sqrt{45}}{2}$. Only positive roots where $(x-1 > 0, x+2 > 0)$ so $(x > 1)$. Both are > 1 (approx 3.854 or -? Actually $(\frac{-1+\sqrt{45}}{2}) \approx (6.708-1)/2 = 2.854$; $(\frac{-1-\sqrt{45}}{2}) \approx -3.854$, invalid). So $(x \approx 2.854)$.

Explanation: Use log sum to product, then solve quadratic, discard extraneous.

26. (Medium, 5 min) Find the natural log: $(\ln(e^3 \cdot e^{2x}))$.

Answer: $(3 + 2x)$.

Explanation: $(\ln(e^3 \cdot e^{2x}) = \ln(e^{3+2x}) = 3+2x)$.

27. (Easy, 2 min) Simplify $(\log_a(a^4))$.

Answer: (4).

Explanation: Definition of logarithm.

28. (Medium, 4 min) Solve for (t) : $(5^{t+1} = 125)$.

Answer: $(125 = 5^3)$. So $(t+1=3)$, $(t=2)$.

Explanation: Write 125 as power of 5.

29. (Hard, 8 min) Solve $(\ln(x) - \ln(x-2) = \ln(3))$.

Answer: $(\ln(\frac{x}{x-2}) = \ln 3)$. So $(\frac{x}{x-2} = 3)$. Solve: $(x = 3(x-2) \implies x = 3x-6 \implies -2x = -6 \implies x = 3)$. Check domain: $(x > 2)$. Accept 3.

Explanation: Combine logs and solve rationally.

30. (Medium, 5 min) Find (x) if $(2^{3x-1} = 7)$.

Answer: $(3x-1 = \log_2 7)$. So $(x = (\log_2 7 + 1)/3)$.

Explanation: Take $\log_2$ of both sides.

Composite and Inverse Functions

31. (Easy, 2 min) Given $(f(x) = 2x+3)$ and $(g(x) = x^2)$, find $((f \circ g)(x))$.

Answer: $((f(g(x)) = 2(x^2) + 3 = 2x^2 + 3)$.

Explanation: Apply (g) then (f) .

32. (Easy, 2 min) Find the inverse of $(f(x) = 3x - 4)$.

Answer: $(f^{-1}(x) = \frac{x+4}{3})$.

Explanation: Swap roles and solve $(y=3x-4)$.

33. (Medium, 4 min) If $(h(x) = \sqrt{5x+1})$, find $(h^{-1}(x))$.

Answer: Solve $(y = \sqrt{5x+1}) \Rightarrow (y^2 = 5x+1) \Rightarrow (x = \frac{y^2-1}{5})$. Thus $(h^{-1}(x) = \frac{x^2-1}{5})$.

Explanation: Swap (x,y) and solve. Domain/range consideration aside.

34. (Medium, 5 min) Compute $((g \circ f)(2))$ for $(f(x) = x^2)$, $(g(x) = \sqrt{x+1})$.

Answer: $(f(2)=4)$, then $(g(4)=\sqrt{5} = \sqrt{5})$.

Explanation: Evaluate inside function then outside.

35. (Hard, 6 min) Verify if $(f(x) = \frac{1}{x})$ is invertible and find $(f^{-1}(x))$.

Answer: Solve $(y = \frac{1}{x})$. Swap: $(x = \frac{1}{y})$. Then $(x(y-1)=1)$, $(xy - x = 1)$, $(xy = x+1)$, $(y = \frac{x+1}{x})$. So $(f^{-1}(x) = \frac{x+1}{x})$. Domain restrictions: $(x \neq 0)$.

Explanation: Invert by swapping variables and solving algebraically.

Trigonometry (Basics and Graphs)

36. **(Easy, 2 min)** Convert (60°) to radians.
Answer: $(\pi/3)$.
Explanation: $(60^\circ) \times (\pi/180) = \pi/3$.
37. **(Easy, 2 min)** Evaluate $(\sin(\pi/6))$.
Answer: $(1/2)$.
Explanation: Known special angle.
38. **(Easy, 2 min)** Solve for $(0 \leq x < 2\pi)$: $(\sin x = 0)$.
Answer: $(x=0, \pi, 2\pi)$. (Actually (2π) is not $<2\pi$, so 0 or π .)
Explanation: Solutions at integer multiples of π .
39. **(Medium, 4 min)** Solve $(\cos(2x) = 0)$ for $(0 \leq x < 2\pi)$.
Answer: $(2x = \pi/2 + n\pi \implies x = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4)$.
Explanation: Use general solution for cosine.
40. **(Medium, 5 min)** Graph one period of $(y = 2\cos(x - \pi/2))$. Describe amplitude, period, and shift.
Answer: Amplitude 2, period (2π) . Phase shift $(\pi/2)$ to right. (Graph: starts at 0 since $(\cos(-\pi/2)=0)$).
Explanation: Standard cosine graph scaled, shifted.
41. **(Medium, 4 min)** If $(\tan(\theta) = -\sqrt{3})$ and (θ) is in QII, find (θ) .
Answer: Reference: $(\sqrt{3})$ corresponds to $(\pi/3)$. In QII, $(\theta = \pi - \pi/3 = 2\pi/3)$.
Explanation: Tan negative means QII or QIV. QII chosen, $\tan = \tan(\pi - \phi)$.
42. **(Hard, 7 min)** Prove the identity: $(1 + \tan^2 x = \sec^2 x)$. (Outline reasoning.)
Answer: This follows from the Pythagorean identity $(1 + \tan^2 x = \sec^2 x)$. Starting with $(\sin^2 + \cos^2 = 1)$, divide by $(\cos^2 x)$ to get $(\tan^2 x + 1 = \sec^2 x)$.
Explanation: Divide basic identity by $(\cos^2 x)$.
43. **(Medium, 4 min)** Solve $(\sin^2 x = \frac{1}{4})$ for $(0 \leq x < 2\pi)$.
Answer: $(\sin x = \pm 1/2)$. Solutions: $(x = \pi/6, 5\pi/6, 7\pi/6, 11\pi/6)$.
Explanation: Use reference angles and quadrants.
44. **(Medium, 5 min)** Evaluate $(\sin(30^\circ)\cos(60^\circ))$.
Answer: $(\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4})$.
Explanation: Use known values.
45. **(Hard, 8 min)** Simplify $(\sin(x + \pi/2))$.
Answer: $(\sin(x + \pi/2) = \cos x)$.
Explanation: Use the co-function identity: $(\sin(\alpha + \pi/2) = \cos \alpha)$.
46. **(Medium, 4 min)** Find the exact value: $(\cos(\frac{5\pi}{3}))$.
Answer: $(1/2)$. (Since $(5\pi/3 = 360^\circ - 60^\circ)$.)
Explanation: Cosine in 4th quadrant equals cos of reference angle.
47. **(Hard, 7 min)** Solve $(2\sin x - 1 = 0)$ on $([0, 2\pi])$.
Answer: $(\sin x = 1/2)$. Solutions: $(x = \pi/6, 5\pi/6)$.
Explanation: Solve and use unit circle.
48. **(Medium, 5 min)** Graph $(y = \tan(x))$ near $(x=0)$. What is its slope at 0?
Answer: Around 0, $(\tan x \approx x)$ (small-angle). Slope $(= \sec^2(0)=1)$.
Explanation: Derivative of tan is $(\sec^2)$, at 0 it's 1.

Trigonometry (Identities and Equations)

49. **(Medium, 4 min)** Prove or simplify: $(\sin^2 x = 1 - \cos^2 x)$.
Answer: From $(\sin^2 x + \cos^2 x = 1)$, rearrange.
Explanation: Fundamental identity rearranged.
50. **(Medium, 5 min)** Solve $(\sin x \cdot \cos x = \frac{1}{2})$ for $(0 \leq x < 2\pi)$.
Answer: Note $(\sin 2x = 2\sin x \cdot \cos x)$. So $(2\sin x \cdot \cos x = 1) \implies (\sin 2x = 1) \implies (2x = \pi/2 + 2\pi k)$. Solutions for $(x: x = \pi/4 + \pi k)$. For $(0 \leq x < 2\pi)$: $(x = \pi/4, 5\pi/4)$.
Explanation: Use double-angle identity.

51. (Hard, 7 min) Simplify $(\tan x + \cot x)$ in terms of $(\sin)$ and $(\cos)$.

Answer: $(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}) = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x}$.

Explanation: Combine fractions and use $(\sin^2 + \cos^2 = 1)$.

52. (Hard, 8 min) Verify the identity: $(\frac{1 - \tan^2 x}{1 + \tan^2 x}) = \cos(2x)$.

Answer: Left side $(= \frac{\cos^2 - \sin^2}{\cos^2 + \sin^2})$. Since $(\cos^2 + \sin^2 = 1)$, this equals $(\cos^2 - \sin^2 = \cos(2x))$.

Explanation: Use $(\tan^2 = \frac{\sin^2}{\cos^2})$ and simplify.

53. (Medium, 5 min) Solve $(\cos(2x) = 0)$ on $([0, 2\pi])$.

Answer: $(2x = \pi/2, 3\pi/2) \Rightarrow (x = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4)$.

Explanation: Solve for $2x$ first, then divide by 2.

54. (Medium, 5 min) Use identity to simplify: $(\sin(x)\cos(x))$.

Answer: $(\frac{1}{2}\sin(2x))$.

Explanation: Product-to-sum identity: $(\sin 2x = 2\sin x \cos x)$.

55. (Hard, 8 min) Expand and simplify: $(\sin(x+y) + \sin(x-y))$.

Answer: Using sum/difference formulas: $(\sin x \cos y + \cos x \sin y + \sin x \cos y - \cos x \sin y = 2\sin x \cos y)$.

Explanation: Combine like terms after expansion.

56. (Hard, 8 min) Prove the identity: $(\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y})$.

Answer: Derive from $(\sin)$ and $(\cos)$ sums: $(\tan(x+y) = \frac{\sin(x+y)}{\cos(x+y)} = \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y})$. Divide numerator and denominator by $(\cos x \cos y)$.

Explanation: Standard derivation from angle-sum formulas.

57. (Medium, 5 min) Solve $(\tan(x) = \sqrt{3})$ for $(0 \leq x < \pi)$.

Answer: $(\tan x = \sqrt{3})$ at $(x = \pi/3, 4\pi/3)$. But in $([0, \pi])$, only $(\pi/3)$.

Explanation: Use reference angle $(\pi/3)$, $\tan$ positive in QI and QIII.

Complex Numbers & Polar Coordinates

58. (Easy, 2 min) Compute $((3+2i) + (1-5i))$.

Answer: $(4 - 3i)$.

Explanation: Add components.

59. (Easy, 2 min) Compute $((4-3i) - (1+2i))$.

Answer: $(3 - 5i)$.

Explanation: Subtract components.

60. (Medium, 4 min) Multiply $((2+3i)(4-i))$.

Answer: $(8 - 2i + 12i - 3i^2 = 8 + 10i + 3 = 11 + 10i)$.

Explanation: Distribute and use $(i^2 = -1)$.

61. (Medium, 5 min) Express $(1 - i)$ in polar form $(re^{i\theta})$.

Answer: Magnitude $(r = \sqrt{1^2 + (-1)^2} = \sqrt{2})$. Argument $(\theta = -\pi/4)$ (or $(7\pi/4)$). So $(\sqrt{2}e^{-i\pi/4})$.

Explanation: Compute length and angle.

62. (Hard, 6 min) Compute $(\frac{1}{2+i})$ in standard $(a+bi)$ form.

Answer: Multiply by conjugate: $(\frac{1}{2+i} \cdot \frac{2-i}{2-i} = \frac{2-i}{4+1} = \frac{2-i}{5} = \frac{2}{5} - \frac{1}{5}i)$.

Explanation: Rationalize denominator.

63. (Hard, 6 min) Solve $(z^2 = -4)$ in complex numbers.

Answer: $(z = 2i)$ or $(z = -2i)$.

Explanation: $(\pm 2i)$.

64. (Medium, 4 min) If $(z = re^{i\theta})$ and $(w = re^{-i\theta})$, show $(z+w)$ is real.

Answer: $(z+w = r(\cos\theta + i\sin\theta) + r(\cos\theta - i\sin\theta) = 2r\cos\theta)$. Real.

Explanation: Imaginary parts cancel.

65. (Medium, 5 min) Convert rectangular $((x,y) = (1, \sqrt{3}))$ to polar $((r,\theta))$.

Answer: $(r = \sqrt{1+3} = 2)$, $(\theta = \tan^{-1}(\sqrt{3}/1) = \pi/3)$.

Explanation: Use $(r = \sqrt{x^2 + y^2})$, $(\tan\theta = y/x)$.

Parametric Equations and Motion

66. (**Easy, 2 min**) For parametric ($x=t+1$, $y=2t$), eliminate (t).
Answer: ($t=x-1$); so ($y = 2(x-1) = 2x-2$).
Explanation: Solve first eq for (t).
67. (**Medium, 4 min**) A particle moves with ($x(t)=3t$), ($y(t)=4t^2$). At ($t=2$), find velocity vector.
Answer: ($v_x = dx/dt = 3$), ($v_y = dy/dt = 8t$). At ($t=2$): ($v=(3,16)$).
Explanation: Differentiate wrt (t).
68. (**Hard, 6 min**) Given ($x=2\cos t$, $y=2\sin t$), show this parametrically describes a circle and find its radius.
Answer: Square & add: ($x^2+y^2 = 4(\cos^2 t + \sin^2 t) = 4$). Radius 2.
Explanation: Use identity ($\cos^2 + \sin^2 = 1$).
69. (**Medium, 5 min**) Eliminate (t): ($x=te^t$, $y=e^t$). (Hint: solve for (t)).
Answer: ($y=e^t$ implies $t=\ln y$). Then ($x = (\ln y) y$). Implicit: ($x = y \ln y$).
Explanation: Use invertible relation for (y). Domain ($y>0$).
70. (**Hard, 8 min**) A projectile has parametric motion ($x=50t$), ($y=50t - 4.9t^2$). Find the time when ($y=0$) (landing).
Answer: Solve ($50t - 4.9t^2 = 0$ implies $t(50 - 4.9t)=0$). Solutions: ($t=0$) (launch) or ($t = 50/4.9 \approx 10.204$) s.
Explanation: Factor out (t).

Vectors and Matrices

71. (**Easy, 2 min**) Compute the dot product of ($\mathbf{a}=(1,2)$) and ($\mathbf{b}=(3,-4)$).
Answer: ($1 \cdot 3 + 2 \cdot (-4) = 3 - 8 = -5$).
Explanation: Sum of products of components.
72. (**Medium, 4 min**) Find the length of vector ($\mathbf{a}=(2,-1,2)$).
Answer: ($\sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4+1+4} = 3$).
Explanation: Use ($|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$).
73. (**Medium, 5 min**) Given ($\mathbf{a}=(1,0,1)$) and ($\mathbf{b}=(2,1,-1)$), find the angle between them.
Answer: Compute dot: ($1 \cdot 2 + 0 \cdot 1 + 1 \cdot (-1) = 2 - 1 = 1$). Magnitudes: ($|\mathbf{a}|=\sqrt{2}$, $|\mathbf{b}|=\sqrt{6}$). ($\cos \theta = \frac{1}{\sqrt{2}\sqrt{6}} = \frac{1}{\sqrt{12}} = \frac{\sqrt{3}}{6}$). So ($\theta = \cos^{-1}(\frac{\sqrt{3}}{6})$).
Explanation: Use ($\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$).
74. (**Easy, 2 min**) Compute ($2 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$).
Answer: ($\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$).
Explanation: Scalar multiplication.
75. (**Medium, 4 min**) Compute ($\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$).
Answer: ($\begin{pmatrix} 1 \cdot 2 + 2 \cdot (-1) \\ 3 \cdot 2 + 4 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$).
Explanation: Matrix times vector.
76. (**Hard, 7 min**) If ($|\mathbf{a}|=3$) and ($|\mathbf{b}|=4$) and the angle between is (60°), find ($\mathbf{a} \cdot \mathbf{b}$).
Answer: ($\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos 60^\circ = 3 \cdot 4 \cdot \frac{1}{2} = 6$).
Explanation: Dot formula: ($\mathbf{a} \cdot \mathbf{b} = ab \cos \theta$).
77. (**Hard, 8 min**) For vectors ($\mathbf{a}=(4,0)$), ($\mathbf{b}=(0,3)$), compute ($\mathbf{a} \times \mathbf{b}$) (2D cross product as scalar).
Answer: In 2D, ($a_x b_y - a_y b_x = 4 \cdot 3 - 0 \cdot 0 = 12$). (Or interpret in 3D as ($12\mathbf{k}$)).
Explanation: Perp product gives scalar in plane.
78. (**Medium, 5 min**) Find the result of ($\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$).
Answer: Multiply: ($\begin{pmatrix} 1 \cdot 0 + 2 \cdot 1 & 1 \cdot 1 + 2 \cdot 0 \\ 3 \cdot 0 + 4 \cdot 1 & 3 \cdot 1 + 4 \cdot 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$).
Explanation: Perform row-by-column multiplication.

Conic Sections and Miscellaneous

79. (**Easy, 2 min**) Identify the conic: ($x^2 + 4y^2 = 4$).
Answer: An ellipse.
Explanation: Two squared terms with same sign, different coefficients.

80. (Medium, 5 min) Put into standard form: $(9x^2 - 4y^2 = 36)$.

Answer: Divide by 36: $(\frac{x^2}{4} - \frac{y^2}{9} = 1)$. Hyperbola.

Explanation: Normalizing to 1.

81. (Medium, 5 min) Sketch or describe $(y = \frac{1}{2}x^2 - x)$. Vertex?

Answer: Vertex at $(x = \frac{-b}{2a} = \frac{1}{1} = 1)$. $(y(1) = 0.5 - 1 = -0.5)$. Parabola opening up.

Explanation: Use vertex formula.

82. (Easy, 2 min) Expand $((x-2)^2 + (y+3)^2 = 25)$.

Answer: $(x^2 - 4x + 4 + y^2 + 6y + 9 = 25)$, so $(x^2 + y^2 - 4x + 6y - 12 = 0)$.

Explanation: Algebraic expansion and simplification.

83. (Medium, 4 min) Compute the derivative of $(f(x) = x^3 - x)$ (for a physics application, etc.).

Answer: $(f'(x) = 3x^2 - 1)$.

Explanation: Power rule.

84. (Easy, 2 min) Average rate of change of $(f(x) = x^2)$ from $(x=1)$ to $(x=4)$.

Answer: $(\frac{f(4)-f(1)}{4-1} = \frac{16-1}{3} = 5)$.

Explanation: Slope formula.

85. (Hard, 8 min) Solve $(\sqrt{x+2} = x - 2)$.

Answer: Square both sides: $(x+2 = x^2 - 4x + 4)$. $(\Rightarrow x^2 - 5x + 2 = 0)$. Solutions: $(x = \frac{5 \pm \sqrt{25-8}}{2} = \frac{5 \pm \sqrt{17}}{2})$. Approx $(x \approx 0.438, 4.562)$. Check original: LHS ≥ 0 requires $(x \geq -2)$. Also RHS ≥ 0 requires $(x \geq 2)$. So only $(x \approx 4.562)$ is valid (since $-0.438 < 2$ fails RHS).

Explanation: Solve algebraically, then check extraneous.

86. (Hard, 8 min) A line passes through $((1,2))$ and is perpendicular to the line $(3x - 4y + 5 = 0)$. Find its equation.

Answer: Slope of given line $(3x-4y+5=0)$ is $(3x+5=4y \Rightarrow y = \frac{3}{4}x + 5/4)$. So slope = $3/4$. Perp. slope = $-4/3$. Using point-slope: $(y-2 = -\frac{4}{3}(x-1))$.

Explanation: Perpendicular slopes are negative reciprocals.

87. (Medium, 5 min) Compute $(\int_0^1 3x^2 dx)$. (Basic calculus review.)

Answer: $(\left[x^3 \right]_0^1 = 1^3 - 0 = 1)$

Explanation: Antiderivative of $(3x^2)$ is (x^3) .

88. (Medium, 4 min) Simplify the expression: $(\frac{d}{dx}(5x^3 - 2x))$.

Answer: $(15x^2 - 2)$.

Explanation: Differentiate term-by-term.

89. (Hard, 7 min) Evaluate the sum of the infinite geometric series $(5 + 15 + 45 + \dots)$.

Answer: Ratio $(r=3)$. Actually diverges $(|r|>1)$, no sum. If meant $5 + 15 + 45$ is not convergent. If meant $5 + 15 + 45 + \dots$ infinite, diverges. If misprint, skip.

Explanation: Because common ratio >1 , no finite sum.

90. (Medium, 4 min) If $(y = \log_{10}(100x))$, what is (dy/dx) ?

Answer: $(y = \log_{10}100 + \log_{10}x = 2 + \log_{10}x)$. So $(dy/dx = 1/(x \ln 10))$.

Explanation: Use derivative of $(\log_{10} x)$.

Additional Practice (Mixed Concepts)

91. (Easy, 2 min) Convert 0.75 into a fraction.

Answer: $(3/4)$.

Explanation: $0.75 = 75/100 = 3/4$.

92. (Easy, 2 min) Evaluate $(\lfloor 2.7 \rfloor)$ (floor function).

Answer: 2.

Explanation: Floor of 2.7 is 2.

93. (Medium, 4 min) If $(f(x)=ax+b)$ and $(f(1)=4, f(3)=10)$, find (a,b) .

Answer: Two equations: $(a+b=4, 3a+b=10)$. Subtract: $(2a=6)$, so $(a=3)$. Then $(b=1)$.

Explanation: Solve linear system.

94. (Medium, 5 min) Compute $\binom{5}{2}$.

Answer: (10).

Explanation: $\frac{5!}{2!3!}=10$.

95. (Hard, 6 min) A function $f(x)$ has values $f(1)=2$, $f(2)=5$, $f(3)=10$. Estimate $f'(2)$ using a symmetric difference quotient.

Answer: Approx $f'(2) \approx \frac{f(3)-f(1)}{3-1} = \frac{f(10-2)}{2} = 4$.

Explanation: $\frac{f(2+h) - f(2-h)}{2h}$ with $h=1$.

96. (Medium, 4 min) Simplify the Boolean expression: $(A \wedge \neg A) \vee B$.

Answer: (B). (Because $A \wedge \neg A = \text{false}$.)

Explanation: A and not A is always false; OR with B yields B.

97. (Hard, 7 min) In a right triangle, one angle is 30° . If the hypotenuse is 10, find the side opposite the 30° angle.

Answer: Side = $10 \sin(30^\circ) = 10 \cdot \frac{1}{2} = 5$.

Explanation: Sine of 30° is $\frac{1}{2}$.

98. (Hard, 7 min) Simplify $\frac{\sin^2(x)}{\cos(x)}$.

Answer: $2\sin(x)\cos(x) = \sin(2x)$.

Explanation: Use chain rule: $2\sin(x)\cos(x)$.

99. (Medium, 4 min) Evaluate the determinant $\begin{vmatrix} 2 & 3 \\ 4 & -1 \end{vmatrix}$.

Answer: $2 \cdot (-1) - 3 \cdot 4 = -2 - 12 = -14$.

Explanation: $ad - bc$.

100. (Medium, 5 min) If a function $g(x)$ passes through $(2,7)$ and has slope (4) there, write its linear approximation at $x=2$.

Answer: $L(x) = 7 + 4(x-2)$.

Explanation: $L(x)=g(2) + g'(2)(x-2)$.

(This completes 100 problems covering varied topics.)

Answer Keys and Solutions

All answers above include brief explanations. Each solution is concise to illustrate the method. The problems are labeled by difficulty (**Easy**, **Medium**, **Hard**) and estimated time. When practicing, start with the easier problems and pace yourself—each easy problem should take ~2–3 min, medium ~4–6 min, and hard ~7–10 min. Use these solutions for self-checking.

Printable Layouts

For easy printing, format schedules and problem sets as HTML tables and lists with simple CSS. For example:

```

<style>
  body { font-size: 12pt; font-family: sans-serif; margin: 1em; }
  table { border: 1px solid #333; border-collapse: collapse; width: 100%; }
  th, td { border: 1px solid #333; padding: 6px; text-align: center; }
  .break { background: #f0f0f0; }
  @media print {
    body { margin: 0.5in; }
    table { page-break-inside: avoid; }
  }
</style>

<table>
  <tr><th>Time</th><th>Day 1</th><th>Day 2</th><th>Day 3</th></tr>
  <tr><td>9:00–10:00</td><td>Quadratic/Polynomial Block</td><td>Exponential Block</td><td>Complex/Polar Block</td></tr>
  <tr class="break"><td>10:00–10:15</td><td colspan="3">Break</td></tr>
  <!-- more rows -->
</table>

<ul>
  <li>1. (Easy) Find roots of  $x^2-5x+6=0$ . <strong>Answer:</strong> 2, 3.
    <em>(Expl: factor  $(x-2)(x-3)=0$ .)</em></li>
  <!-- more items -->
</ul>

```

This snippet uses a small font and borders for clarity. For problem sets, use numbered HTML lists `<ol>`. Ensure page breaks don't split problems awkwardly (use CSS `page-break-inside: avoid` if needed). Adjust margins and font sizes for legibility in print view.

Study Techniques and Resources

- **Active Recall:** Regularly practice problems instead of passive reading. Pulling answers “out of your head” strengthens memory 【26†L61-L63】.
- **Spaced Practice:** Review material multiple times. E.g., revisit each topic on multiple days rather than cramming in one session.
- **Time-Blocking:** Follow the schedule above and stick to planned blocks. This creates structure and reduces procrastination 【17†L156-L163】.
- **Pomodoro Breaks:** Work 25–50 minutes, then 5–10 minute breaks 【17†L156-L163】. Longer 15–20 min breaks after 2–3 blocks help reset focus 【21†L202-L210】 【40†L203-L208】. Avoid distractions (like social media) during breaks.
- **Review Sessions:** Use the end-of-day review blocks for quick concept checks and sample questions. Reflect on what was learned each day.
- **Teach or Explain:** Try explaining key ideas aloud or to a study partner; this solidifies understanding.

Resources: Authoritative sources such as the AP Precalculus Course Description outline important topics and skills 【5†L159-L167】. Online platforms like Khan Academy, College Board’s AP resources, and math forums (with caution) provide additional practice. High-quality textbooks or open textbooks (e.g. UTSA Precalculus) can supplement.

References

- AP College Board, *AP Precalculus Course and Exam Description* (Fall 2023) – Unit topics and weightings 【5†L159-L167】.
- Schoolhouse.world, *How to Create a Study Schedule* – Time-blocking and Pomodoro recommendations 【17†L156-L163】.
- Cornell University Health, *Study Breaks & Stress-Busters* – Benefits of taking breaks (5–60 min) 【40†L203-L208】.
- ++Tutors Blog, *Study Break Routines* – 20-min break after 90-min study improves retention 【21†L202-L210】.
- P. Agarwal, *RetrievalPractice.org, What is Retrieval Practice?* – Active recall boosts learning 【26†L61-L63】.